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Swing

Student Information System

<u>Section A: GUI Architecture Overview</u>

Overall Structure: The application follows a **single-window desktop GUI architecture** built using Java Swing. The design separates:

- **User Interface (UI components)**
- **Event handling logic**
- **Data storage and file persistence**

The program runs on the **Event Dispatch Thread (EDT)** using: `SwingUtilities.invokeLater(() -> new StudentInfoApp());`

Main Window

- Implemented using JFrame
- Title: *Student Information System*
- Layout: BorderLayout

Input Components

- JTextField → Student Name input
- JComboBox → Course selection
- JButton → Save, Clear
- JTextField → Search field
- JButton → Search Name

Display Components

- JTable → Displays student records
- DefaultTableModel → Stores table data
- JScrollPane → Enables scrolling
- JMenuBar → File & Help menu

Data Storage Structure

- **Runtime Storage**: DefaultTableModel
- **Persistent Storage**: students.txt
- Format: Name,Course

Control Flow

1. User enters data.
2. Clicks **Save**.
3. ActionListener triggers handleSave().
4. Data is validated.
5. Added to tableModel.
6. Saved to file.
7. Table updates automatically.

UI → Event → Logic → Model → File → UI refresh

<u>Section B: Swing Component Selection</u>

JFrame

- **Purpose:** Main application window.
- **Why appropriate:** Top-level container for Swing apps.
- **Limitation:** Only one root window; complex apps need multiple frames or dialogs.

JTextField

- **Purpose:** Input student name and search text.
- **Why appropriate:** Designed for single-line input.

- **Limitation:** Does not support multi-line input.

JComboBox

- **Purpose:** Select course.
- **Why appropriate:** Restricts input to predefined options.
- **Limitation:** Limited scalability if options become very large.

JButton

- **Purpose:** Trigger Save, Clear, Search actions.
- **Why appropriate:** Standard event-trigger component.
- **Limitation:** Requires event listener for functionality.

JTable

- **Purpose:** Display structured student data.
- **Why appropriate:** Tabular data display with sorting support.
- **Limitation:** Complex customization requires additional configuration.

<u>Section C: Event Handling and Listener Logic</u>

Events Generated

- Save button click
- Clear button click
- Search button click
- Menu item click (Exit, About)

ActionListner Processing

Lambda expressions are used: saveButton.addActionListener(e -> handleSave());

When triggered:

- Data is retrieved
- Validation occurs
- Model updates
- File saved

Data Transfer Process

1. nameField.getText()
2. courseBox.getSelectedItem()
3. Data added via tableModel.addRow()
4. Persisted via saveToFile()
5. Display updated automatically

<u>Section D: Validation and Error Handling</u>

```
if (name.isEmpty()) {
   JOptionPane.showMessageDialog(frame, "Name is required!", "Error",
JOptionPane.ERROR_MESSAGE);
   return;
}
```

On Invalid Input

- Error dialog appears.
- Data is NOT added.
- File is NOT modified.

Why Validation is Necessary

- Prevents empty records
- Maintains data consistency
- Avoids corrupt file entries
- Improves user experience

Section E: Layout Management Reasoning

Layout Managers Used

- BorderLayout → Main frame
- GridBagLayout → Input panel
- FlowLayout → Search panel & button panel

Why These Layouts?

BorderLayout

- Separates top controls and center table cleanly.

GridBagLayout

- Flexible grid positioning for labels and fields.
- Allows resizing adaptability.

FlowLayout

- Simple left-aligned search components.

Resizing Behavior

- JTable expands with window.
- Top panels remain structured.

This ensures responsive UI without distortion.

Section F: Data Handling and Persistence

Data Storage During Runtime

Stored in DefaultTableModel.

File Persistence

File: students.txt

Format: John,Math
Sara,Science

Save Process

- Loops through table rows
- Writes comma-separated values

Load Process

- Reads line-by-line
- Splits using .split(",")
- Adds rows to table model

UI and File Consistency

- After every Save → File overwritten with current table state.
- Prevents mismatch between UI and file.

Section G: What-If Design Analysis

1. What problems arise if all logic is written inside event handlers?

- Code becomes messy and unmaintainable.
- Difficult to debug.
- Hard to reuse logic.
- Violates separation of concerns principle.

Your project improves this by using separate methods like:

- handleSave()
- saveToFile()
- loadFromFile()

4. What happens if long tasks run on the Event Dispatch Thread?

- UI freezes.
- Buttons stop responding.
- Window becomes unresponsive.

Swing is single-threaded; heavy tasks should use SwingWorker.

Section H: Individual Extension Explanation

Implemented Extension: Search Functionality with TableRowSorter

I used: TableRowSorter<DefaultTableModel>

RowFilter.regexFilter("(?i)" + text)

Impact on UI Flow

- User enters search text.
- Table dynamically filters.
- Case-insensitive search.
- Improves usability significantly.

This makes the application interactive and scalable.

Section I: Debugging and Learning Reflection

Issue Encountered

If file did not exist initially, application could fail while loading.

Diagnosis

Used: if (!file.exists()) return;

Learning Improvement

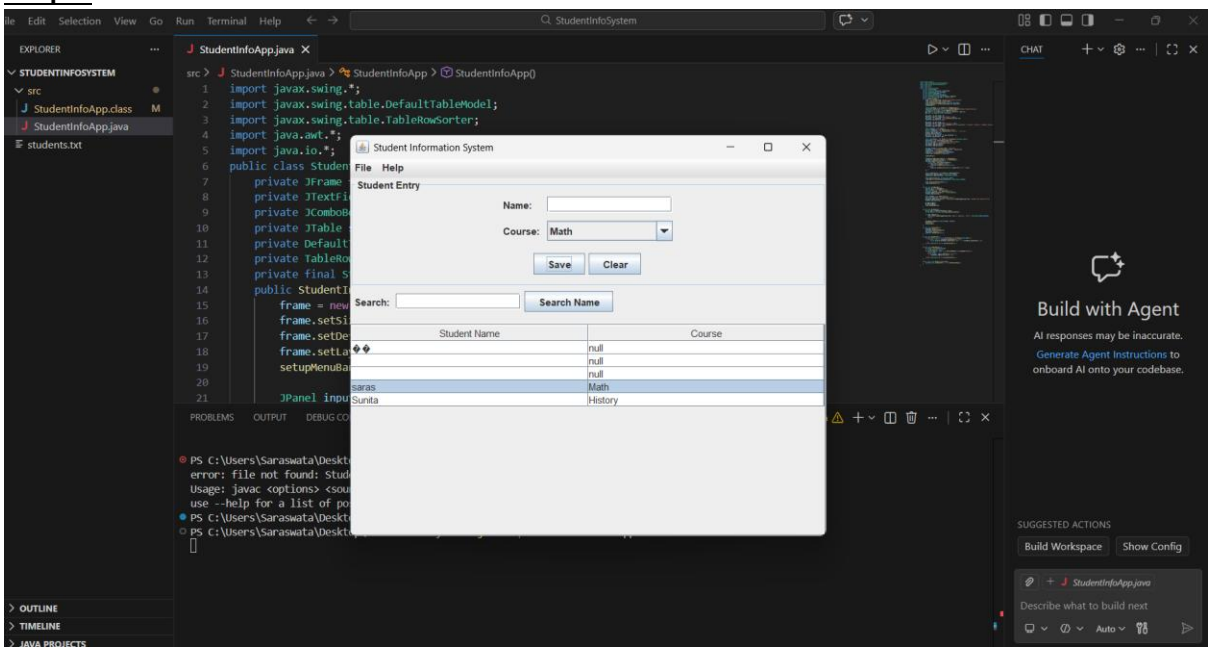
- Understood importance of file existence checks.
- Learned difference between runtime and persistent storage.
- Improved understanding of MVC-like separation in Swing.

Section J: AI Usage Disclosure

AI assistance was used for:

- Structuring the project report.

Usability was validated by checking:

- Proper error dialogues

Output